Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skillsthrough emerging technologies.

Session 2026-2027

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name of Student

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Session	**2026-27 (ODD)**	**Course Name**	**PE-V : Lab : Blockchain Technology**
Semester	**7**	**Course Code**	**23IOT1748**
Roll No	**61**	**Name of Student**	**Sheikh Wasimuddin**

Practical Number	**02**
Course Outcome	**1. Demonstrate basics of distributed systems and cryptographic primitives used in blockchain.** **2. Outline various research aspects for development of blockchain.** **3. Make use of different security protocols to build secure blockchain applications.** **4. Analyze smart contracts and Hyperledger Fabric used in blockchain. 5. Determine the performance of applications developed with Bitcoin and Ethereum blockchain.**
Aim	Implement SHA-256 Hashing and Analyze the Avalanche Effect and generate and verify Hash Values of Text Files Using SHA-256.
Theory (100 words)	SHA-512 (Secure Hash Algorithm-512) is a cryptographic hash function that converts any input data, such as text, files, or messages, into a fixed **512-bit (64-byte)** hash value, which is typically represented as a **128-character hexadecimal string**. It belongs to the **SHA-2 family** of cryptographic algorithms developed by the **National Security Agency (NSA)** and is widely used for ensuring data integrity, digital signatures, secure communication, password protection, and blockchain technology. SHA-512 is designed to provide a higher level of security than SHA-256 by generating a longer hash value, making it more resistant to collision and brute-force attacks. It is a one-way function, meaning the original input cannot be recovered from the generated hash. The same input always produces the same hash value, while even the smallest change in the input results in a completely different hash due to the **Avalanche Effect**, a key property of cryptographic hash functions. This effect ensures that changing even a single character or bit in the input significantly alters the output hash, making it impossible to predict the relationship between similar inputs and their corresponding hashes. SHA-512 is commonly used for **hash generation and verification**. During hash generation, the algorithm computes a unique 512-bit hash for a file, message, or text. During verification, the hash of the received data is generated again and compared with the original hash value. If both hash values are identical, the data is confirmed to be authentic and unmodified. If

<table>
<tr><td></td><td>the hash values differ, it indicates that the file has been altered, corrupted, or tampered with. Due to its strong security features, SHA-512 is widely used in applications such as file integrity verification, digital signatures, SSL/TLS certificates, blockchain systems, secure password hashing, software distribution verification, and data authentication.</td></tr>
<tr><td>Procedure and Execution

(100 Words)</td><td>

mplementation Steps

Experiment 2(A): SHA-512 Hash Generation Using Online Tool

1. Open **https://emn178.github.io/online-tools/sha512.html** in a web browser.
2. Ensure the **Input Encoding** is set to **UTF-8**.
3. Type the desired text into the **Input** box.
4. Click the **Hash** button.
5. Observe the generated **128-character SHA-512 hash** in the **Output** section.
6. Copy and save the generated hash value.
7. Modify the input text by changing or adding a single character.
8. Generate the hash again.
9. Compare the two hash values. The new hash will be completely different, demonstrating the **Avalanche Effect**.

Experiment 2(B): SHA-512 Hash Generation Using Windows Command Prompt

1. Open **Notepad** and create a text file.
2. Enter some text in the file and save it.
3. Open **Command Prompt**.
4. Navigate to the folder containing the text file using the **cd** command.

cd C:\Users\student\Desktop

5. Generate the SHA-512 hash of the text file using the Windows **certutil** command.

certutil -hashfile "BCT_Lab 2.txt" SHA512

6. Note the generated **128-character SHA-512 hash** value.
7. Modify the contents of the text file by changing even a single character (for example, change **Hello** to **Hello!** or add one letter) and save the file.
8. Generate the SHA-512 hash again using the same command.

certutil -hashfile "BCT_Lab 2.txt" SHA512

9. Compare the two hash values.
    - If the file is unchanged, both hashes will be identical.
    - If even one character is changed, the new hash will be completely different, demonstrating the **Avalanche Effect**.

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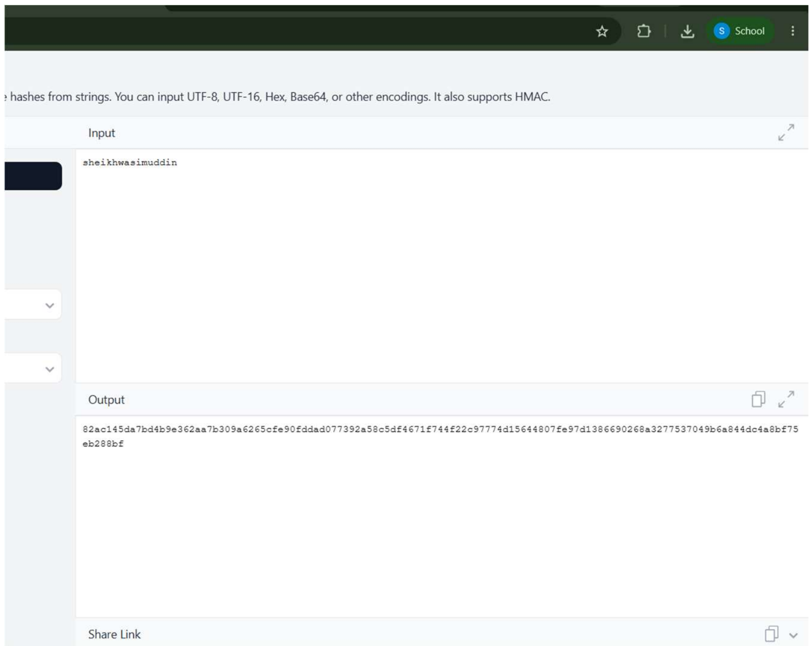
	Stepwise Screenshots with steps:



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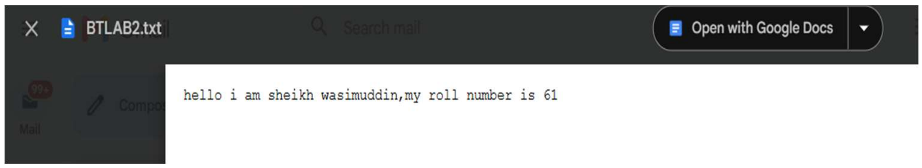

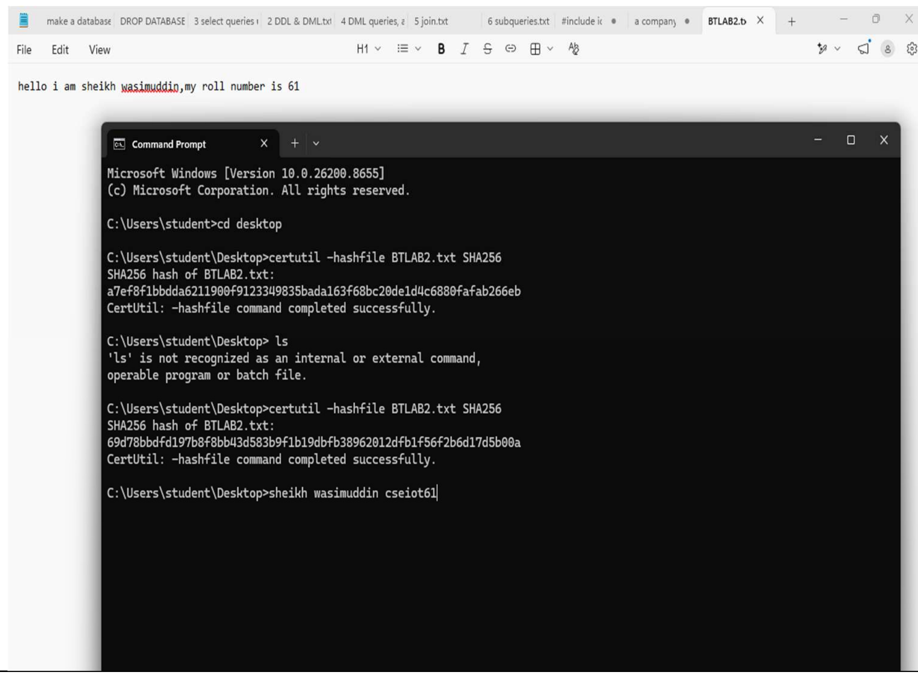

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<tr><td>Conclusion</td><td>The SHA-512 hashing algorithm was successfully implemented to generate and verify hash values for text files. Hash values were generated using both an online SHA-512 tool and the Windows certutil command. It was observed that even a small change in the input text produced a completely different hash value, demonstrating the Avalanche Effect. The experiment also showed that identical files produce the same hash, while modified files produce different hashes, making SHA-512 an effective method for ensuring data integrity, authenticity, and enhanced security due to its larger 512-bit hash output.</td></tr>
<tr><td>Date</td><td>14/07/2026</td></tr>
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